

Pharmacology Lecture

Neuromuscular Blocking Agents
(NMBAs)

Part 1: Learning Objectives

NMBAs:

- Review the molecular steps of neurotransmission at the NMJ.
- Differentiate the mechanism and clinical use of depolarizing vs. non-depolarizing agents.
- Identify key pharmacokinetic factors impacting NMBA choice (e.g., organ elimination).
- Recognize the adverse effects and management of malignant hyperthermia.
- Compare traditional (Neostigmine) and targeted (Sugammadex) reversal agents.

This module focuses on agents used to achieve skeletal muscle paralysis during surgery and mechanical ventilation.

Clinical Uses of Neuromuscular Blockade

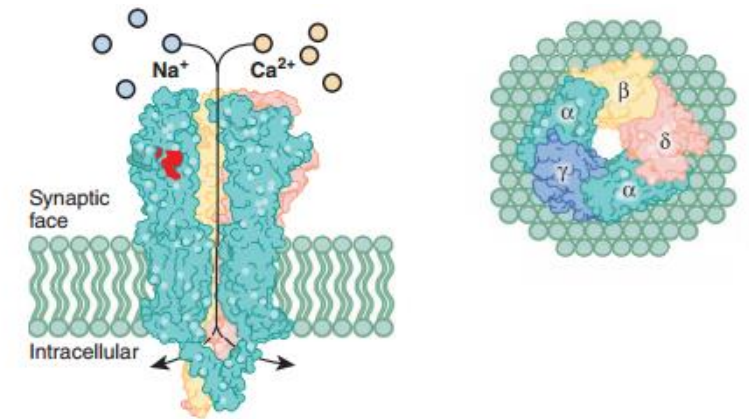
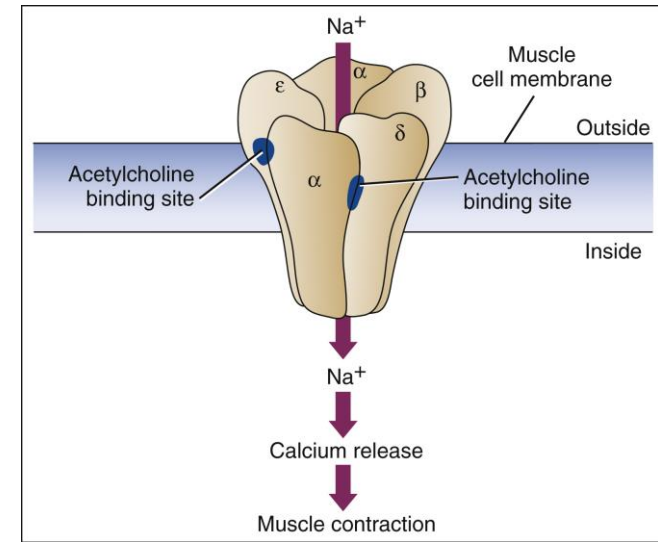
Surgical Procedures(Anesthesia) Critical Care/Emergency

- Optimizing surgical field exposure (abdominal, thoracic, orthopedic).
- Preventing sudden patient movement during delicate procedures (e.g., neurosurgery, ophthalmology).
- Facilitating robotic surgery.
- Facilitating Rapid Sequence Intubation (RSI) to secure the airway.
- Managing severe tetanus or status epilepticus (where muscle contraction poses risk).
- Controlling muscle activity during mechanical ventilation in severe ARDS.

The Target: Nicotinic Acetylcholine Receptor (nAChR)

Structure

- A pentameric ligand-gated ion channel
- Adult nAChR composition: two α_1 , one β_1 , one δ , and one ϵ subunit
- highly concentrated at the motor endplate.
- Two ACh molecules must bind, one to each α subunit, to open the channel.



NMJ Physiology: Pre-Synaptic Events

- Action potential arrives at the nerve terminal.
 - Depolarization opens voltage-gated Ca^{2+} channels.
 - Calcium influx initiates the fusion of synaptic vesicles containing ACh.
 - SNARE complex proteins (Synaptobrevin, Syntaxin, SNAP-25) facilitate vesicle docking and fusion.
 - ACh is released into the synaptic cleft.
-

NMJ Physiology: Post-Synaptic Activation

- ACh diffuses across the cleft and binds to the two alpha subunits of the nAChR.
 - This causes a conformational change, opening the central pore.
 - Ions: Na^+ and Ca^{2+} rush into the cell; K^+ leaves the cell. Net effect is depolarization.
 - The resulting End Plate Potential (EPP) reaches threshold to fire a muscle action potential.
 - Muscle action potential spreads via T-tubules, releasing Ca^{2+} from the Sarcoplasmic Reticulum → Contraction.
-

Termination: Acetylcholinesterase (AChE)

Function

- Rapidly hydrolyzes ACh into choline and acetate.
- Ensures quick termination of the EPP, allowing the muscle fiber to repolarize and be ready for the next impulse.
- Located in the synaptic cleft's basement membrane.

Target for Reversal

AChE is the target for the reversal of non-depolarizing block (e.g., Neostigmine) by increasing endogenous ACh concentration.

Classification & Mechanism of NMBAs

Depolarizing vs. Non-Depolarizing Agents

Classification of NMBA's: The Two Groups

Class	Mechanism at nAChR	Prototype	Effect on Endplate	Reversal Strategy
Depolarizing	Agonist (mimics ACh)	Succinylcholine	Sustained Depolarization	
Non-Depolarizing	Competitive Antagonist	Rocuronium	Prevents Depolarization	Antagonist or AChE Inhibitor

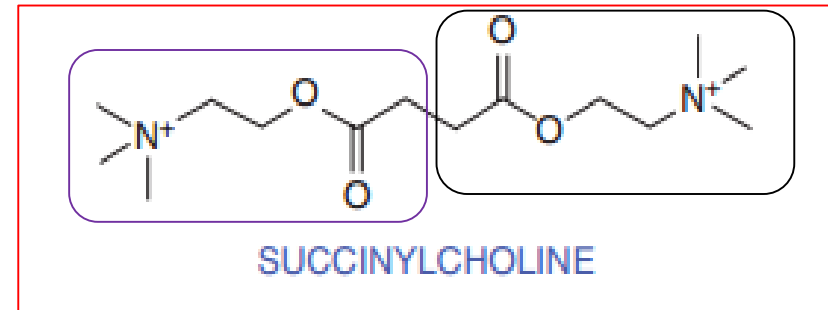
Succinylcholine: Chemical Structure

Structure

- Chemically, it is two Acetylcholine molecules linked end-to-end.
- allows it to bind to the nAChR and act as an agonist.

Clinical Use

- The only clinically used depolarizing blocker.
- Indicated for procedures requiring ultra-short duration and rapid onset (e.g., RSI).



Succinylcholine: Phase I Block (Depolarizing)

Mechanism

- binds and activates nAChR, causing the muscle cell to depolarize.
- **Unlike ACh, it is not rapidly hydrolyzed by AChE.**
- The sustained depolarization keeps the voltage-gated Na⁺ channels surrounding the endplate in an inactive (closed/refractory) state.
- The muscle cannot fire another action potential, leading to **flaccid paralysis**.

Clinical Manifestation

The initial burst of depolarization causes fasciculations (visible muscle twitching) before paralysis sets in.

Succinylcholine: Phase II Block (Desensitizing)

- Occurs with high doses or prolonged infusion of Succinylcholine.
 - The endplate gradually repolarizes, **but the nAChR becomes desensitized and resistant to activation by ACh.**
 - The mechanism becomes functionally similar to a non-depolarizing block.
 - Clinical relevance: If paralysis is unexpectedly prolonged, Phase II block may be developing.
-

Onset and Duration

- Onset: Ultra-fast, typically < 60 seconds. (Ideal for RSI).
- Duration: Ultra-short, typically 5-10 minutes.

Metabolism

- Primary route: Hydrolyzed by Plasma **Butyrylcholinesterase** (also known as **Pseudocholinesterase**) in the plasma, not by AChE at the NMJ.
- The short duration is due to **rapid diffusion away from the NMJ** and quick breakdown in the bloodstream.

Atypical Butyrylcholinesterase (BChE)

Genetic Variation

- Autosomal recessive condition affecting the structure or quantity of BChE.
- The enzyme is functionally impaired, leading to **slowed metabolism of Succinylcholine**.
- Patients experience a **prolonged blockade** (sometimes hours) after a standard dose.

Succinylcholine: Major Adverse Effects

Hyperkalemia

- Depolarization of the muscle membrane causes a **release of K⁺ from the muscle cell**, usually leading to a safe 0.5 mEq/L rise.
- Absolute Contraindication: In conditions involving **upregulation of extra-junctional nAChR** (risk of massive K⁺ release):
 - Burns (> 24 hours post-injury)
 - Massive trauma or crush injury
 - Spinal cord injury (paraplegia/quadriplegia)
 - Severe intra-abdominal infection

Malignant Hyperthermia: Uncontrolled Calcium Release

- Triggering agent (Succinylcholine or Volatile) binds RyR1.
- This causes massive, uncontrolled efflux of Ca^{2+} from the SR into the myoplasm.
- High intracellular Ca^{2+} causes sustained muscle contraction (rigidity).
- Massive ATP consumption leads to heat production (hyperthermia) and metabolic crisis.

1. Hypermetabolism

Massive ATP use, increased O_2 consumption, and CO_2 production.

2. Heat Production

Uncontrolled skeletal muscle activity generates excessive body heat (hyperthermia).

3. Cellular Damage

Lactic acidosis, hyperkalemia, and rhabdomyolysis.

Malignant Hyperthermia: Clinical Manifestation

Early Signs

- Massive increase in End-Tidal CO₂ (ETCO₂): Often the earliest and most sensitive sign of hypermetabolism.
- Tachycardia and Arrhythmias.
- Masseter spasm or generalized muscle rigidity.

Late Signs

- Rapidly rising Core Temperature (Hyperthermia, > 40C).
 - Metabolic Acidosis (low pH).
 - Rhabdomyolysis (high CK and myoglobinuria).
-

Malignant Hyperthermia: Treatment Protocol

Pharmacological Intervention

Dantrolene Sodium: [The definitive antidote](#).

- Mechanism: Acts directly on the skeletal muscle RyR1 receptor to reduce Ca^{2+} efflux from the SR. This uncouples excitation-contraction.
- Dosing: IV administration, 2.5 mg/kg repeated until control is achieved.

Supportive Measures

- Discontinue all triggering agents.
- Hyperventilate with 100% O_2 (to eliminate CO_2).
- Cool the patient aggressively (IV cold saline, lavage).
- Treat acidosis (Sodium Bicarbonate).
- Treat hyperkalemia and arrhythmias (Lidocaine, Glucose/Insulin).

Other Adverse Effects

Cardiovascular Effects

- Bradycardia: Especially in children or with second dose in adults, due to stimulation of cardiac muscarinic receptors (ACh mimetic).
- Tachycardia and hypertension (less common).

Pressure Increases

- Intraocular Pressure (IOP): Contraindicated in open-globe injury.
 - Intra gastric Pressure: Fasciculations can increase pressure, theoretically increasing aspiration risk.
-

Non-Depolarizing Blocking Agents (NDMBAs)

Antagonism and Organ Elimination

NDMBAs: Competitive Antagonism

Mechanism

- NDMBAs are highly charged, bulky quaternary ammonium compounds.
 - They bind reversibly to the alpha subunits of the nAChR but do not activate the receptor.
 - They competitively inhibit ACh from binding, thus preventing channel opening and depolarization.
 - Result: Flaccid paralysis without prior fasciculations.
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NDMBA Classification by Chemical Structure

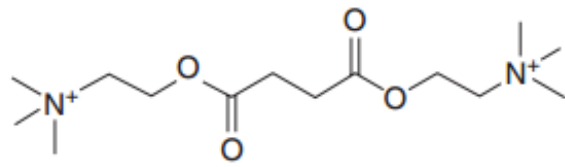
1. Aminosteroidal Agents

- Structure: Possess a steroid ring structure.
- Examples: Rocuronium, Vecuronium, Pancuronium.
- Elimination: Primarily Hepatic and Renal clearance.
- Reversal: Highly effective reversal by Sugammadex

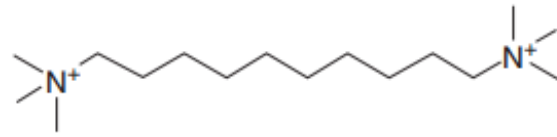
2. Benzylisoquinolinium Agents

- Structure: Based on a benzylisoquinoline nucleus.
- Examples: Cisatracurium, Atracurium.
- Elimination: Mostly organ-independent (Hoffman elimination).
- Reversal: Reversal by Sugammadex is ineffective.

Depolarizing Neuromuscular Blockers

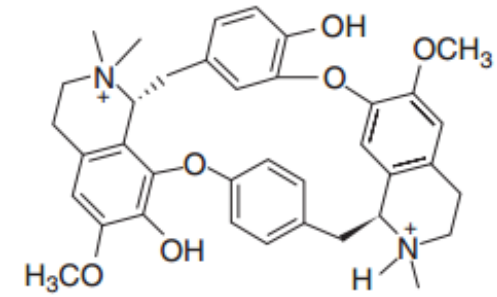


SUCCINYLCHOLINE

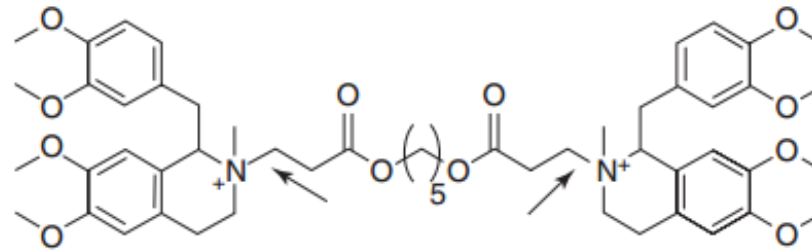


DECAMETHONIUM

Benzylisoquinoline Competitive Neuromuscular Blockers



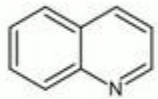
TUBOCURARINE



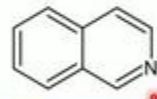
ATRACURIUM

arrows: cleavage sites for Hofmann elimination

Quinoline and Isoquinoline

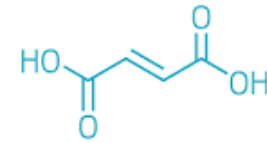
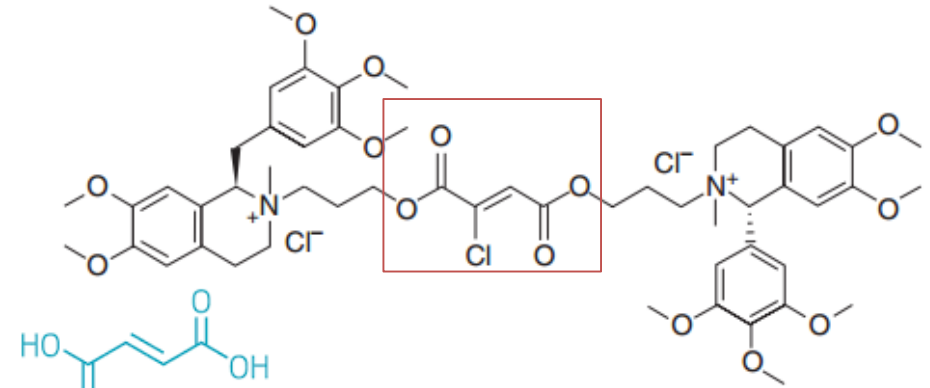


Quinoline



Isoquinoline

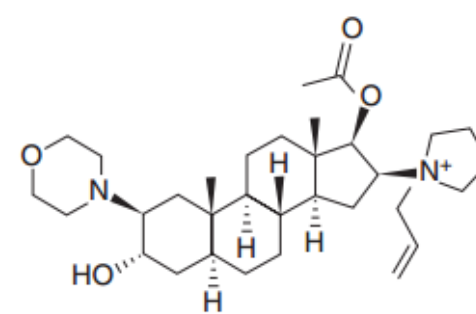
Mixed-Onium Chlorofumarate Competitive Neuromuscular Blockers



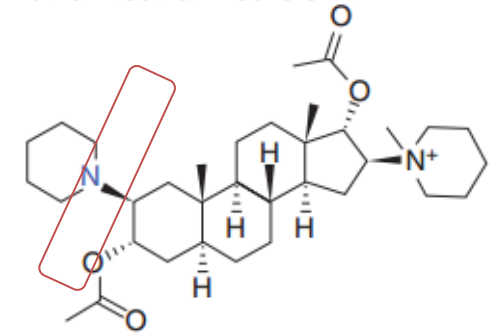
Fumaric acid

GANTACURIUM

Aminosteroid Competitive Neuromuscular Blockers



ROCURONIUM



VECURIUM

PANCURONIUM: addition of CH₃ at N

Pharmacokinetic Variables: Onset and Duration

Agent	Class	Onset (mins)	Duration (mins)	Primary Elimination
Rocuronium	Aminosteroid	1-2(Fast)	30-45 (Intermediate)	Hepatic (>70%)
Vecuronium	Aminosteroid	2-4 (Intermediate)	30-45 (Intermediate)	Hepatic/Renal
Cisatracurium	Benzylisoquinoline	2-4(Intermediate)	30-45 (Intermediate)	Hoffman (Organ-Independent)
Pancuronium	Aminosteroid	3-5(Slow)	90-120(Long)	Renal

AGENT Chemical Class <i>Type of action</i>	ONSET (min) ^a	DURATION (min) ^a	MODE OF ELIMINATION
Ultrashort and short duration			
Succinylcholine DCE, <i>depolarizing</i>	0.8–1.4	6–11	Hydrolysis by plasma cholinesterases
Gantacurium ^c MOCE, <i>competitive</i>	1–2	5–10	Cysteine adduction, ester hydrolysis
Mivacurium BIQ, <i>competitive</i>	2–3	15–21	Hydrolysis by plasma cholinesterases
Intermediate duration			
Vecuronium AS, <i>competitive</i>	2–3	25–40	Hepatic and renal elimination
Atracurium BIQ, <i>competitive</i>	3	45	Hofmann elimination; ester hydrolysis
Rocuronium AS, <i>competitive</i>	0.5–2	36–73	Hepatic elimination
Cisatracurium BIQ, <i>competitive</i>	2–8	45–90	Hofmann elimination; renal elimination

AGENT Chemical Class <i>Type of action</i>	ONSET (min) ^a	DURATION (min) ^a	MODE OF ELIMINATION
Long duration			
Pipecuronium ^b AS, <i>competitive</i>	3–6	30–90	Renal elimination; hepatic metabolism/ clearance
D-Tubocurarine ^b CBI, <i>competitive</i>	6	80	Renal and hepatic elimination
Pancuronium AS, <i>competitive</i>	3–4	85–100	Renal and hepatic elimination
Metocurine ^b BIQ, <i>competitive</i>	4	110	Renal elimination
Doxacurium ^b BIQ, <i>competitive</i>	4–8	120	Renal elimination

^aAs achieved from dose ranges in Table 11–4.

^bNot commercially available in the U.S.

^cGantacurium is in investigational status.

Abbreviations: AS, aminosteroid; BIQ, benzylisoquinoline; CBI (natural alkaloid), cyclic benzylisoquinoline; DCE, dicholine ester; MOCE, assymetric mixed-onium chlorofumarate.

NDMBAs: Elimination Pathways

Hepatic/Renal (Rocuronium, Vecuronium, Pancuronium)

- ❖ Dependent on liver function (metabolism) and kidney function (excretion of parent drug/metabolites).

Risk:

- ❖ Prolonged blockade in patients with hepatic or renal failure.

Hoffman Elimination (Cisatracurium, Atracurium)

- ❖ Spontaneous, non-enzymatic chemical breakdown in plasma and tissues.
- ❖ Rate is dependent on pH and temperature.

Benefit:

- ❖ Drug of choice in patients with severe organ failure (liver or kidney).

Atracurium Metabolism

- ❖ Atracurium's breakdown produces Laudanosine, a CNS stimulant that can cause seizures (though rarely at clinical doses).

Note:

- ❖ Cisatracurium produces far less Laudanosine and is therefore preferred.

Factors Affecting NMB Response

Potentiation (↑ effect)

- ❖ Aminoglycosides, magnesium, volatile anesthetics
- ❖ Hypothermia, acidosis, myasthenia gravis
- ❖ Renal/hepatic impairment (most agents)

Resistance (↓ effect):

- ❖ Hyperthermia, alkalosis
- ❖ Burn patients (after 24-72 hours)
- ❖ Upregulated nAChRs
- ❖ Anticonvulsants (chronic use)

Reversal Agents

- ❖ Neostigmine, pyridostigmine, edrophonium
- ❖ **Sugammadex**: Cyclodextrin that encapsulates rocuronium/vecuronium
- ❖ rapid reversal (reverse deep blockade)

**Anticholinesterases + anticholinergic
OR sugammadex**

Reversal of NMBs: Cholinesterase Inhibitors

Neostigmine, Pyridostigmine

❖ Mechanism:

Inhibit Acetylcholinesterase ->

Accumulation of ACh -> Displaces the competitive blocker.

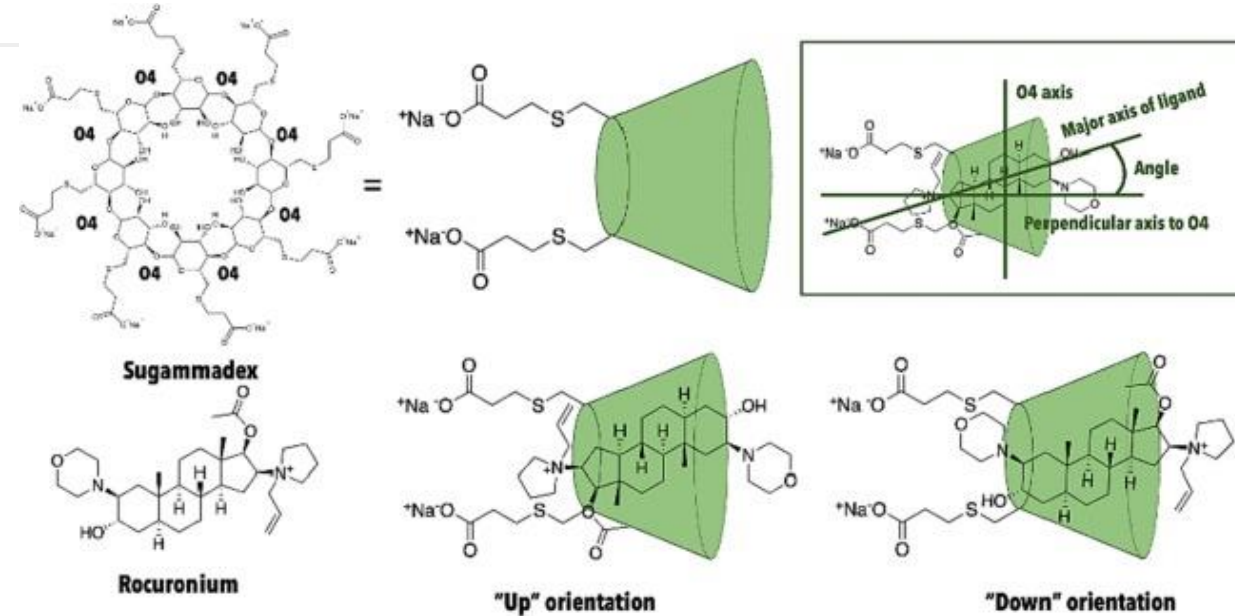
Side Effects (Muscarinic)

- ❖ Increased ACh affects the heart and gut.
- ❖ Causes bradycardia, salivation, diarrhea.
- ❖ Must administer with an **Antimuscarinic** (Glycopyrrolate or Atropine).

Reversal of NMBs: Sugammadex

Selective Relaxant Binding Agent

- ❖ A cyclodextrin ring that encapsulates the NMB molecule (specifically Rocuronium and Vecuronium) in the plasma, rendering it inactive.



Encapsulation or chelation

- ❖ **Benefit:** Rapid reversal, no muscarinic side effects.
- ❖ **Limitation:** Only works for steroidal NMBs (Roc/Vec). Expensive.



Spasmolytics

Treating Spasticity and Spasms

Definitions: Spasticity vs. Spasm

Spasticity

- ❖ Upper Motor Neuron lesion (CP, MS, Stroke).
- ❖ Hyper-excitability of stretch reflex.
- ❖ Sustained contraction.

Acute Muscle Spasm

- ❖ Peripheral musculoskeletal injury.
- ❖ Involuntary tension due to inflammation or trauma.

Drugs for Chronic Spasticity

- Baclofen
- Diazepam (Benzodiazepines)
- Tizanidine
- Dantrolene

Baclofen

- **Mechanism: GABAB agonist.**
- Increases K⁺ conductance -> Hyperpolarization.
- **Reduces release of excitatory transmitters** (Glutamate) in the spinal cord.

**Clinical Use: Multiple Sclerosis,
Spinal Cord Injury.**

Adverse Effects

Sedation, weakness. Abrupt withdrawal can cause seizures and hallucinations (Must taper).

Diazepam (Valium)

Mechanism

- Potentiates GABA-A receptors (Cl⁻ channel). Increases presynaptic inhibition in the spinal cord.
- **Use:** Effective for spasticity but highly sedating.
- **Drawback:** Tolerance and physical dependence potential.

Tizanidine

Mechanism: Alpha-2 Adrenergic Agonist (Central). Congener of Clonidine.

Effect: Reduces spasticity with **less muscle weakness** than Baclofen or Diazepam.

Side Effects: Drowsiness, Hypotension, Dry mouth, Hepatotoxicity (Monitor LFTs).

Dantrolene

Mechanism (Unique)

Directly acts on skeletal muscle. Blocks **Ryanodine Receptor (RyR1)** channel (major calcium (Ca^{2+}) release channel). Prevents **Calcium release** from Sarcoplasmic Reticulum.

Indications: Malignant Hyperthermia (IV), Chronic Spasticity (PO).

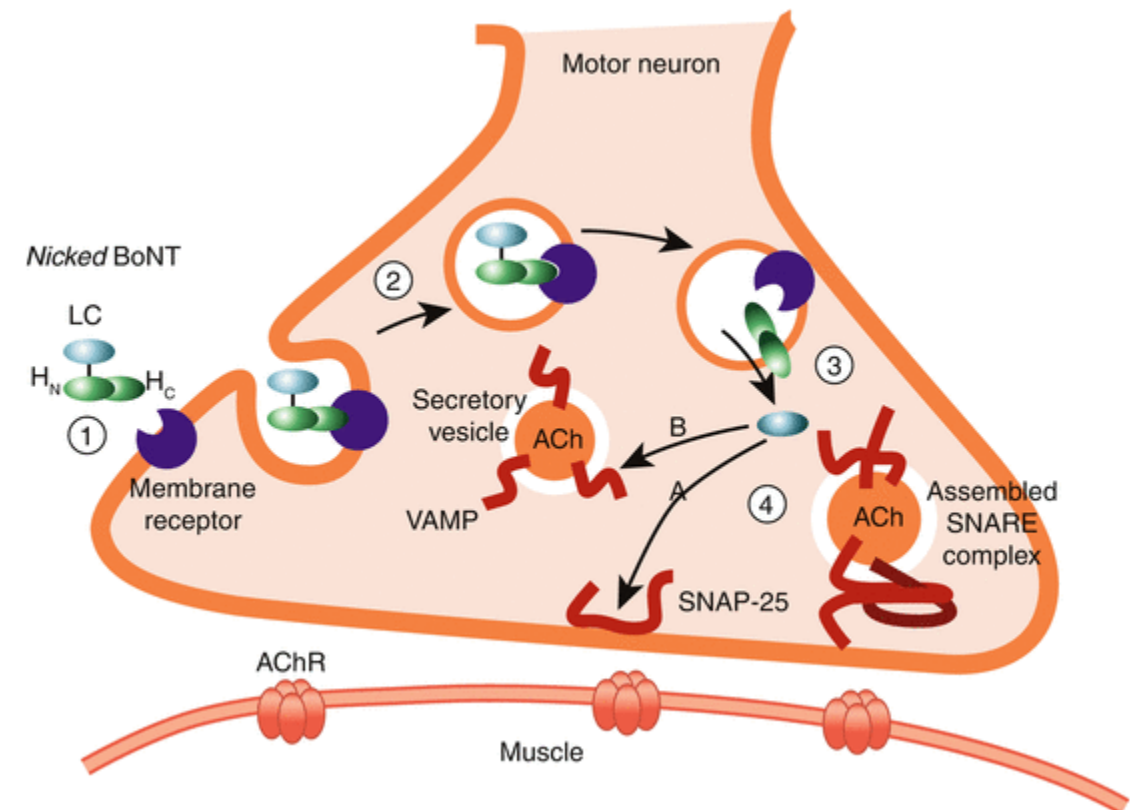
Side Effect: Generalized muscle weakness, Hepatotoxicity.

Botulinum Toxin (Botox)

Mechanism: Cleaves SNARE proteins (SNAP-25). Prevents vesicular release of ACh at the NMJ.

Use: Localized spasticity (e.g., Cerebral Palsy), Dystonias, Cosmetic.

Duration: Effects last 3-4 months (until nerve terminals resprout).



Drugs for Acute Muscle Spasm

Used for short-term relief (2-3 weeks) alongside rest and PT.

- ❖ **Cyclobenzaprine:** Structurally related to TCAs. Strong antimuscarinic effects (Dry mouth, sedation). **5HT₂ receptor antagonist**
- ❖ **Carisoprodol:** Metabolized to Meprobamate (anxiolytic). **Abuse potential (Schedule IV).**
- ❖ **Methocarbamol:** Central depression. Less sedating.

Spasmolytics Summary

Drug	Target	Key Pearl
Baclofen	GABA-B	Avoid abrupt withdrawal.
Diazepam	GABA-A	Sedation/Addiction risk.
Tizanidine	Alpha-2	Less weakness, check BP.
Dantrolene	Muscle (RyR1)	Tx for Malignant Hyperthermia.
Cyclobenzaprine	Brainstem	Anticholinergic side effects.



NSAIDs & Acetaminophen

Inflammation and Pain Management

The Arachidonic Acid Pathway

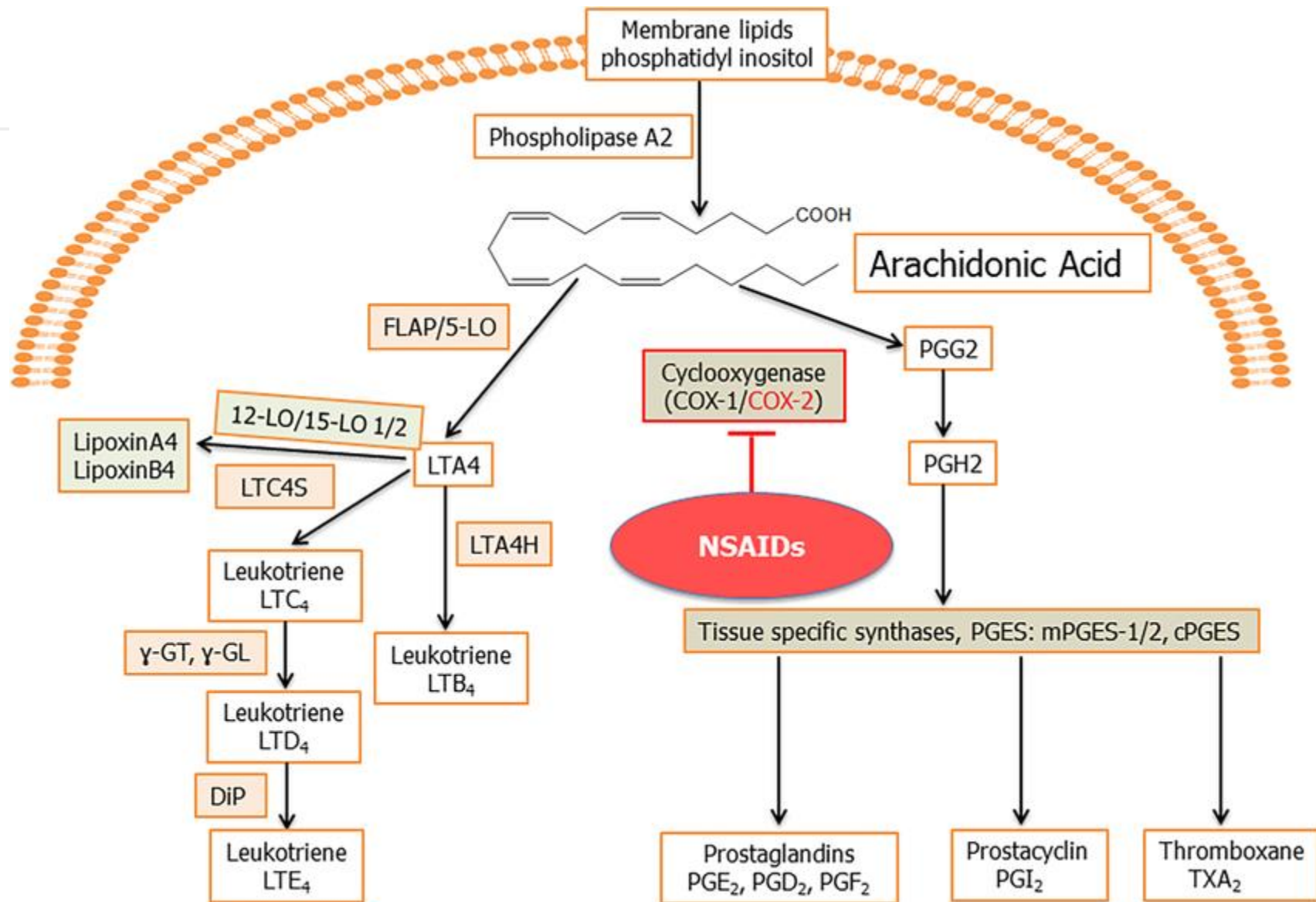
Membrane Phospholipids → (Phospholipase A2) → Arachidonic Acid.

COX Pathway

Produces Prostaglandins (Pain, Inflammation, Fever) and Thromboxanes (Clotting).

LOX Pathway

Produces Leukotrienes (Bronchoconstriction).



COX-1 vs. COX-2

COX-1 (Constitutive)

"Housekeeping." Maintains gastric lining (PGE₂), renal blood flow, platelet aggregation (TXA₂).

COX-2 (Inducible)

Induced at [sites of inflammation](#).
Mediates pain, fever, and inflammation. Also constitutive in kidney/endothelium.

NSAIDs: Mechanism of Action

Inhibition of Cyclooxygenase (COX)

Inhibits the conversion of Arachidonic Acid to Prostaglandins and Thromboxanes.

Effects: Analgesic, Antipyretic, Anti-inflammatory.

Aspirin (Acetylsalicylic Acid)

Unique Mechanism: Irreversible acetylation of COX-1 and COX-2.

- ❖ **Platelets:** Permanently inhibits TXA2 production (platelets have no nucleus to regenerate COX). Antiplatelet effect lasts 7-10 days (life of platelet).
- ❖ **Dose:** Low dose (81mg) for cardio-protection. High dose for anti-inflammatory.

Aspirin: Adverse Effects

- ❖ **GI:** Ulcers, bleeding (loss of protective PG).
- ❖ **Respiratory:** "Aspirin Asthma" (Shunt to Leukotrienes).
- ❖ **Reye's Syndrome:** Hepatic encephalopathy in children with viral infections. **Contraindicated in kids.**
- ❖ **Salicylism:** Tinnitus, vertigo, respiratory alkalosis (overdose).

Non-Selective NSAIDs

Ibuprofen, Naproxen, Indomethacin, Ketorolac

Reversible inhibition of COX-1 and COX-2.

Indomethacin: Very potent. Used for Gout and closure of Patent Ductus Arteriosus (PDA).

Ketorolac: Potent analgesic (opioid level). Max use 5 days (renal toxicity).

Naproxen: Longer half-life (BID dosing). Considered safest for CV risk among non-aspirin NSAIDs.

NSAID Adverse Effects: Gastrointestinal

Mechanism: Inhibition of COX-1 reduces Prostaglandin E2 and I2 (which stimulate mucus/bicarb secretion and reduce acid).

Management

Take with food. Co-administer PPI (Omeprazole) or Misoprostol (PG analog) in high-risk patients.

NSAID Adverse Effects: Renal

Mechanism: Prostaglandins maintain dilation of the Afferent Arteriole.

NSAIDs cause Afferent constriction -> Reduced GFR -> Renal failure.

"Triple Whammy"

Avoid combination of: NSAID + ACE Inhibitor + Diuretic. Extremely high risk of Acute Kidney Injury.

Selective COX-2 Inhibitors (Celecoxib)

Designed to spare COX-1 (protect stomach) while inhibiting COX-2 (inflammation).

Benefits

Lower risk of GI ulcers. No effect on platelet aggregation (bleeding time normal).

Risks (Black Box)

Increased risk of Thrombosis (MI/Stroke). Imbalance between Prostacyclin (vasodilator, inhibited) and Thromboxane (clotting, uninhibited).

Acetaminophen (Paracetamol)

Mechanism

Weak inhibitor of COX in peripheral tissues. Inhibits COX-3 (variant) in CNS.

- ❖ **Effects:** Analgesic, Antipyretic. **NOT Anti-inflammatory.**
- ❖ **Safety:** No GI irritation, no renal toxicity, no platelet effect. Safe in kids (no Reye's).

Acetaminophen Metabolism & Toxicity

Normal metabolism:

Glucuronidation/Sulfation (95%).

Minor pathway (CYP2E1): produces toxic **NAPQI**.

NAPQI is detoxified by **Glutathione**.

Overdose

Glutathione depleted -> NAPQI
accumulates -> Centrilobular
Hepatic Necrosis.

Treatment of Acetaminophen Toxicity

N-Acetylcysteine (NAC)

Mechanism: Acts as a sulfhydryl group donor, regenerating Glutathione to neutralize NAPQI.

- Most effective if given within 8-10 hours of ingestion.
- Nomogram used to determine need for treatment based on serum levels.

Opioids in Musculoskeletal Pain (Brief)

Reserved for severe pain (e.g., Trauma, Post-op).

- **Mechanism:** Mu-receptor agonists.
- **Examples:** Morphine, Oxycodone, Hydrocodone.
- **Risks:** Respiratory depression, Constipation, Addiction/Dependence.
- **Strategy:** Use "Multimodal Analgesia" (NSAID + Tylenol + Muscle Relaxant) to spare opioids.

Review: Analgesic Ladder

1. **Mild:** Acetaminophen, NSAIDs.
2. **Moderate:** Combination (Acetaminophen + Codeine/Tramadol).
3. **Severe:** Strong Opioids (Morphine, Fentanyl).
4. **Adjuvant:** Muscle relaxants, Antidepressants, Anticonvulsants.



Rheumatoid Arthritis

Disease-Modifying Antirheumatic Drugs (DMARDs)

RA Pathophysiology & Treatment Goals

Autoimmune destruction of synovial joints. Mediated by TNF-alpha, IL-1, IL-6.

NSAIDs/Steroids

Bridge therapy. Control symptoms
but do **not** stop disease progression
or bone erosion.

DMARDs

Slow/halt disease progression.
Prevent joint deformity. Effect takes
weeks to months.

Disease-Modifying Antirheumatic Drugs

Methotrexate (MTX)

The Gold Standard first-line DMARD.

Mechanism

Folic acid antagonist. Inhibits Dihydrofolate Reductase (DHFR). Inhibits DNA synthesis in immune cells. Also increases Adenosine (anti-inflammatory).

Methotrexate Toxicity

Common

Nausea, mucosal ulcers (stomatitis). Minimized by Folic Acid supplementation (daily, except MTX day).

Serious

Hepatotoxicity, Bone Marrow Suppression, Pulmonary Fibrosis.

Contraindication

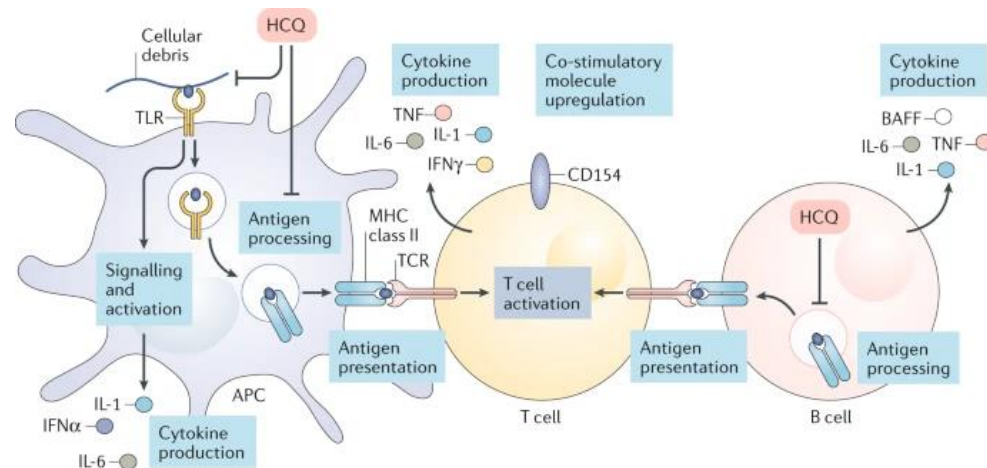
Pregnancy
(Teratogenic).

Hydroxychloroquine

Mechanism: Stabilizes lysosomes, raises pH, disrupts antigen presentation.

Use: Mild RA, SLE (Lupus). Often combined with MTX.

Toxicity: Retinal damage (Retinopathy). Requires annual eye exams. Safe in pregnancy.

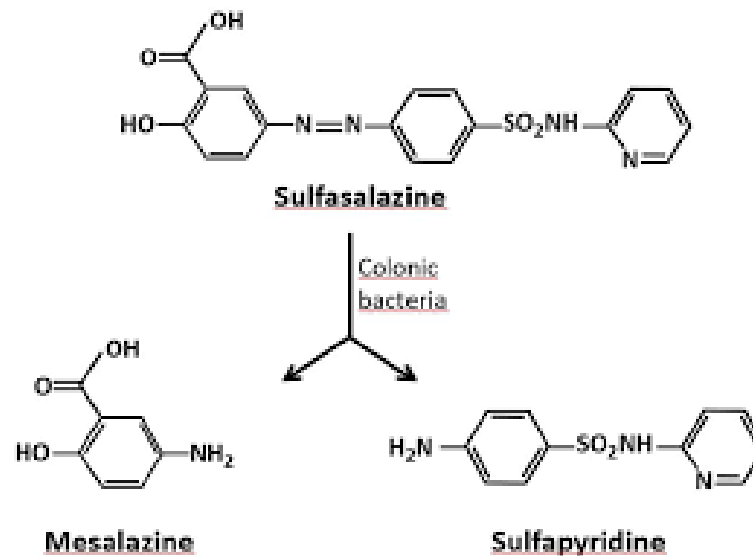


Sulfasalazine

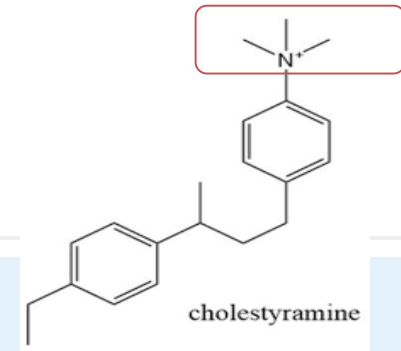
Mechanism: Cleaved by gut bacteria into Sulfapyridine (active for RA) and 5-ASA (active for IBD).

Adverse Effects: GI upset, rash, hemolysis in G6PD deficiency (sulfa drug).

Use: Moderate RA, IBD.



Leflunomide

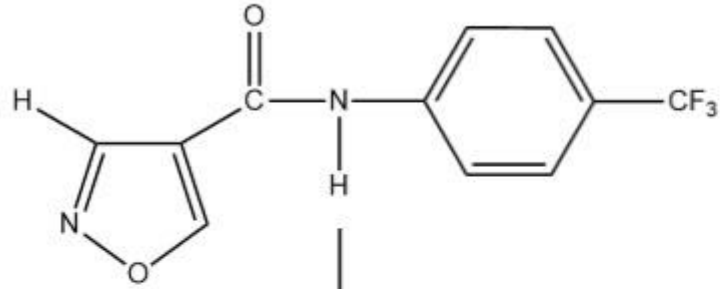


Mechanism

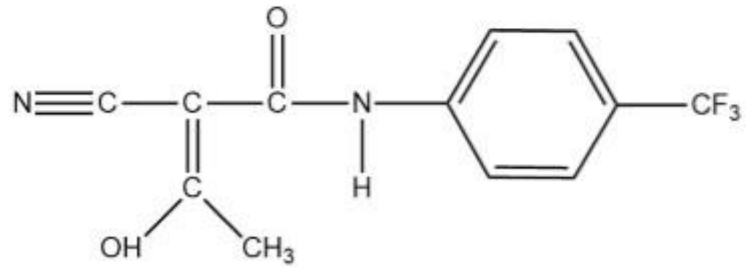
Inhibits Dihydroorotate Dehydrogenase (DHODH). Blocks **Pyrimidine** synthesis. Arrests T-cell proliferation.

- **Side Effects:** Diarrhea, Hepatotoxicity, Teratogenicity (It is hepatotoxic and fetal injury when administered to pregnant women).
- **Washout:** Has a very long half-life. Use **Cholestyramine** to washout if pregnancy desired.
 - Empirical use in the treatment of polyomavirus nephropathy seen in immunosuppressed renal transplant recipients

Leflunomide



Teriflunomide
(Active Metabolite)



Inhibits

T-lymphocyte activation

De novo pathway

DHODH

Pyrimidine pools

RNA and DNA for T-lymphocyte proliferation



Biologic DMARDs: TNF-Alpha Inhibitors

Used when non-biologics (MTX) fail.

Drug	Type
Infliximab	Chimeric Monoclonal Antibody (IV)
Adalimumab	Fully Human Monoclonal Antibody (SC)
Etanercept	Recombinant Receptor Fusion Protein (SC)

Safety Concerns with TNF Inhibitors

Infection Risk

High risk of reactivation of **Latent TB** and Hepatitis B. Patients must be screened (PPD/Quantiferon) before starting.

- Risk of Lymphoma.
- Exacerbation of Heart Failure.
- Injection site reactions.

Other Biologic Agents

- **Rituximab:** Anti-CD20 (Depletes B-cells).
- **Abatacept:** Blocks T-cell co-stimulation (binds CD80/86).
- **Tocilizumab:** IL-6 Receptor antagonist.
- **Anakinra:** IL-1 Receptor antagonist (rarely used in RA now).

Combination Therapy in RA

Monotherapy is often insufficient.

- **Triple Therapy:** Methotrexate + Sulfasalazine + Hydroxychloroquine.
- **Biologic Combo:** Methotrexate + TNF Inhibitor.
- **Contraindication:** NEVER combine two Biologics (e.g., TNF inh + Abatacept). Infection risk is too high.

Glucocorticoids in Musculoskeletal Disease

Prednisone, Methylprednisolone

- **Mechanism:** Inhibit Phospholipase A2. Downregulate pro-inflammatory genes.
- **Use:** Acute flares, bridge therapy. Intra-articular injections.
- **Long-term Risk:** Osteoporosis, Muscle wasting, Adrenal suppression, Cushing's syndrome.

Osteoarthritis (OA) Management

Non-inflammatory (mostly). DMARDs are **NOT** effective.

- 1 Acetaminophen (First line - though efficacy debated).
- 2 Topical NSAIDs (Diclofenac gel).
- 3 Oral NSAIDs (if no contraindications).
- 4 Intra-articular Steroids.
- 5 Surgery.

DMARDs Summary

Drug	Toxicity	Notes
Methotrexate	Liver, Lung, Bone Marrow	Give Folate.
Hydroxychloroquine	Retina	Safe in Pregnancy.
Sulfasalazine	Sulfa allergy, G6PD	Safe in Pregnancy.
TNF Inhibitors	TB Reactivation	Screen TB first.



Gout & Hyperuricemia

Crystal Arthropathy Management

Pathophysiology of Gout

Deposition of Monosodium Urate (MSU) crystals in joints.

Hyperuricemia: High uric acid (>6.8 mg/dL) due to Overproduction (10%) or Underexcretion (90%).

Purine Metabolism: Purines -> Hypoxanthine -> Xanthine -> Uric Acid (via **Xanthine Oxidase**).

Treatment Strategy

Acute Attack

Goal:

- ✓ Stop inflammation and pain.
- ✓ NSAIDs
- ✓ Colchicine
- ✓ Steroids

Chronic (Prophylactic)

Goal: Lower serum Uric Acid (<6.0 mg/dL)

- ✓ Xanthine Oxidase Inhibitors
- ✓ Uricosurics
- ✓ Uricase

Acute Gout: NSAIDs

- **Indomethacin:** Traditionally favored.
- **Naproxen/Ibuprofen:** High doses needed.
- **Avoid Aspirin:** Low doses can inhibit uric acid excretion (competes at OAT transporter).

Acute Gout: Colchicine

Mechanism

Binds to tubulin. **Inhibits microtubule polymerization**. Prevents neutrophil migration (chemotaxis) and phagocytosis of crystals.

Timing: Must start within 24-36 hours of flare onset.

Side Effects: Severe diarrhea (dose-limiting), nausea.

Chronic Gout: Allopurinol

First Line Urate Lowering Therapy (ULT).

Mechanism

Competitively **inhibits Xanthine Oxidase** (reduced form of the XO).

Reduces production of uric acid.

Precipitation of Acute Gout

Starting ULT mobilizes crystals. Must **co-administer Colchicine or NSAID** for first 3-6 months.

Allopurinol: Adverse Effects & Interaction

- ✓ **Hypersensitivity:** Rash. Rare but fatal Steven-Johnson Syndrome (Screen HLA-B*5801 in Asians).
- ✓ **Drug Interaction:** Azathioprine/6-MP. These are metabolized by Xanthine Oxidase. Allopurinol increases their levels to toxic range (**Bone marrow suppression**). Reduce dose by 75%.

Chronic Gout: Febuxostat

Class: Non-purine **Xanthine Oxidase** (oxidized and reduced forms)

Inhibitor.

Use: Patients intolerant to Allopurinol or **with renal failure** (hepatic

Risk: Higher cardiovascular risk warning compared to Allopurinol.

Elimination: extensively metabolized conjugation via UGT enzymes, and oxidation by CYPs, non-CYP enzymes, via hepatic and renal pathways.

Drug Interactions: contraindicated in patients on mercaptopurine, azathioprine

Chronic Gout: Probenecid

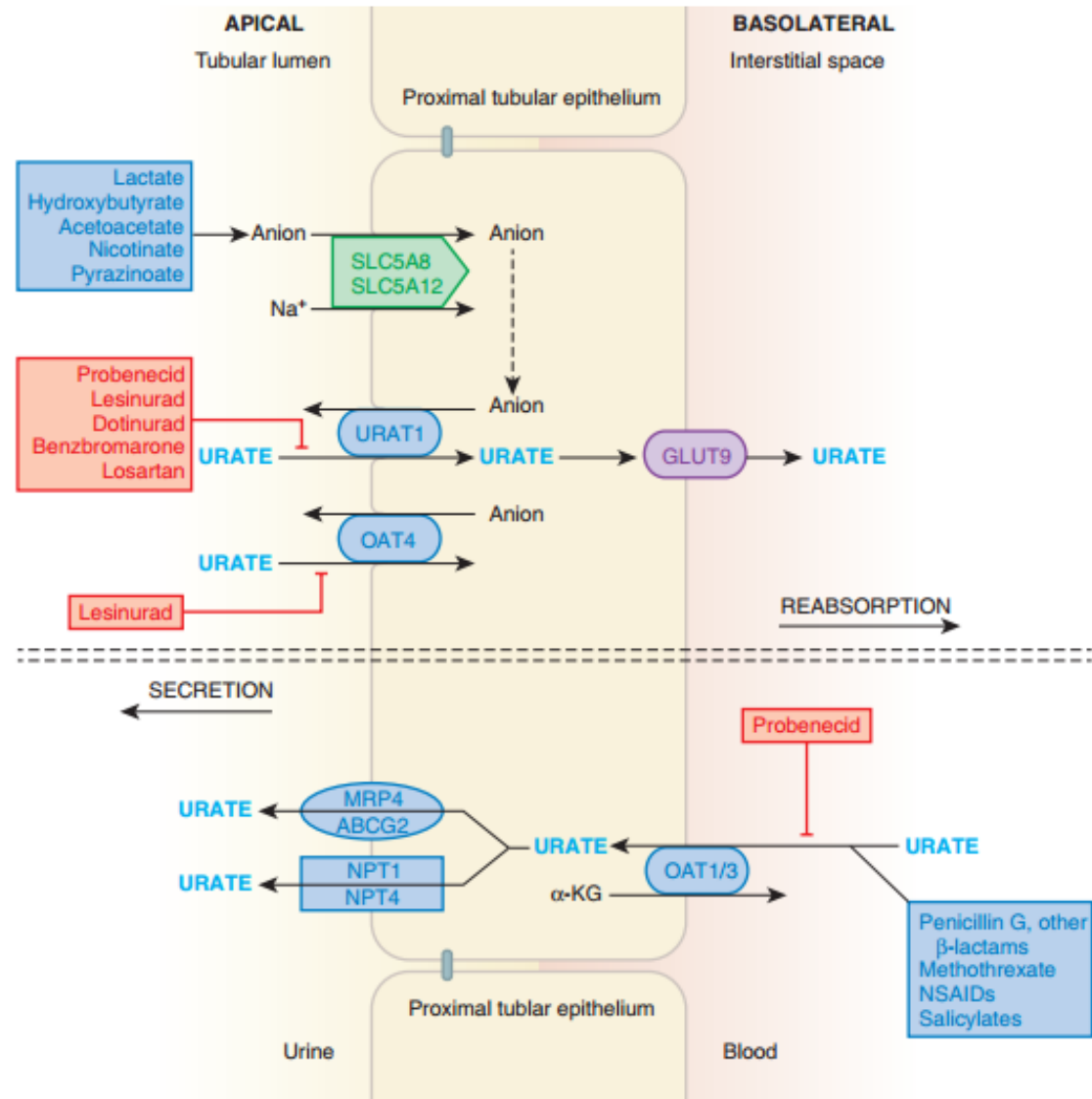
Mechanism (Uricosuric)

Inhibits reabsorption of Uric Acid in the **Proximal Convoluted Tubule** (URAT1 inhibitor). Increases excretion.

Contraindications: Kidney stones, renal insufficiency.

Note: Drink plenty of water to prevent stones.

Chronic Gout: Probenecid



Refractory Gout: Pegloticase

Pegylated uricase (urate oxidase)

Mechanism

Recombinant Uricase (enzyme). Metabolizes Uric Acid into Allantoin (highly water soluble and excreted).

- **Use:** Severe, refractory tophaceous gout. IV infusion.
- **Risk:** Anaphylaxis, Hemolysis in G6PD deficiency.



Bone Mineral Homeostasis

Treating Osteoporosis

Hormonal Regulation of Bone

PTH

- Increases Serum Calcium.
- Stimulates osteoclasts (indirectly).
- Increases renal Ca reabsorption.
- Activates Vit D.

Vitamin D

- Increases intestinal absorption of Calcium and Phosphate.
- Promotes mineralization.

Calcitonin

- Lowers Serum Calcium.
- Inhibits osteoclasts. (Minor role in adults).

Calcium and Vitamin D Supplementation

- **Calcium Carbonate:** Needs acid for absorption (take with meals).
- **Calcium Citrate:** Acid-independent (good for elderly/PPI users).
- **Vitamin D3 (Cholecalciferol):** Preferred supplement. Activated in liver (25-OH) then kidney (1,25-OH).

Indication: Foundation of all osteoporosis therapy.

Bisphosphonates: Mechanism

Alendronate, Risedronate, Zoledronic Acid

Pyrophosphate Analogs

Bind to hydroxyapatite in bone. Taken up by Osteoclasts. Inhibit Farnesyl Pyrophosphate Synthase -> **Osteoclast Apoptosis.**

Result: Reduced bone resorption.

Bisphosphonates: Pharmacokinetics

- **Absorption:** Extremely poor ($<1\%$).
- Must take on empty stomach with water.
- **Distribution:** Binds avidly to bone. Remains for years.

Administration Instructions

- Take with full glass of water.
- Stay upright for 30 minutes.
- Prevents severe **Esophagitis**.

Bisphosphonates: Adverse Effects

- **GI:** Esophagitis, ulcers (Oral). Acute phase reaction (IV).
- **Osteonecrosis of the Jaw (ONJ):** Rare. Associated with high-dose IV use (cancer) and dental procedures.
- **Atypical Femur Fractures:** Associated with long-term use (>5 years).
Drug holiday recommended after 5 years.

Denosumab

Mechanism

Monoclonal antibody against **RANKL**. Mimics OPG. Prevents RANKL from activating osteoclasts.

Use: SubQ every 6 months. Safe in renal failure (unlike bisphosphonates).

Risks: Hypocalcemia, ONJ, Rebound fracture risk upon discontinuation.

SERMs: Raloxifene

Selective Estrogen Receptor Modulator

Agonist at Bone (inhibits resorption). Antagonist at Breast/Uterus.

- **Use:** Postmenopausal osteoporosis + Breast cancer prophylaxis
- **Side Effects:** Hot flashes, DVT/PE risk (like estrogen)

Parathyroid Hormone Analogs (Anabolic)

Teriparatide, Abaloparatide

Concept: Intermittent PTH administration stimulates osteoblasts more than osteoclasts -> New Bone Formation.

Use: Severe osteoporosis. Max use 2 years.

Risk: Osteosarcoma (in rat studies).
Contraindicated in Paget's disease or prior radiation.

Romosozumab

Mechanism

Antibody against **Sclerostin**. Sclerostin normally inhibits bone formation.

Effect: Dual action - Increases bone formation AND decreases resorption.

Warning: Black box for MI/Stroke risk.

Bone Agents Summary

Class	Examples	Primary Action
Bisphosphonates	Alendronate	Antiresorptive
RANKL Inhibitor	Denosumab	Antiresorptive
PTH Analog	Teriparatide	Anabolic
SERM	Raloxifene	Antiresorptive